DS701 Homework 6: k-NN, Naive Bayes, and the SVM Margin

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Name: _____ **BU ID:** _____

How this works. Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it without a computer or an AI assistant. It is **participation-graded**: a serious attempt at every problem earns full credit, and worked solutions are released the day after the deadline. Budget 1–2 hours.

Covers: Lecture 09 (k-Nearest Neighbors, the curse of dimensionality, Naive Bayes), plus one self-contained geometry problem on the SVM margin.

Problem 1: k-NN by hand, and why scale matters

Practices: the k-NN prediction rule, the effect of feature units on Euclidean distance, and the bias–variance role of k.

Eight users of an app are described by x_1 = hours of use per week (0–10) and x_2 = fraction of sessions rated "satisfied" (0–1). Label A = renewed the subscription, B = cancelled.

point	P1	P2	P3	P4	P5	P6	P7	P8
x_1	4	8	6	0	2	7	10	1
x_2	0.2	0.9	0.7	0.2	0.0	0.1	1.0	0.5
label	A	A	A	A	B	B	B	B

A new user has $\mathbf{q} = (4,\ 0.1)$. Use Euclidean distance; since only the *ordering* of distances matters, you may work with squared distances throughout.

(a) Compute the squared distance from $\mathbf{q}$ to each of the eight points. Give the $k = 1$ and the $k = 3$ predictions (majority vote).

point	P1	P2	P3	P4	P5	P6	P7	P8
d^2	_____	_____	_____	_____	_____	_____	_____	_____

$k = 1$ prediction: _____ $k = 3$ prediction: _____

(b) A colleague points out that x_2 is a fraction while x_1 is in hours, and re-expresses x_2 as a score out of 10 (multiply the x_2 column and q_2 by 10). Recompute the squared distances, and give the new $k = 1$ and $k = 3$ predictions. Which prediction changed, and why exactly — which point(s) entered or left the neighbourhood? In one sentence, what does this say about running k-NN on raw features?

New $k = 1$ prediction: _____ new $k = 3$ prediction: _____

(c) What does $k = 8$ predict for *any* query in this dataset, and why? Using the vocabulary of Lecture 07 (bias/variance), explain what moving k from 1 toward 8 does to the decision boundary and to the two kinds of error, and say how k should be chosen in practice.

$k = 8$ predicts: _____

Problem 2: The curse of dimensionality, in numbers

Practices: quantifying "nearest is meaningless" with two short calculations.

Suppose N points are spread uniformly in the unit hypercube $[0,1]^d$. To find the k nearest neighbours of a query, k-NN needs a small box around the query that contains a fraction $f = k/N$ of the data.

(a) Show that a hypercube of side length s inside $[0,1]^d$ captures a fraction s^d of the data, so the side length needed to capture fraction f is $s = f^{1/d}$. (Two or three lines.)

(b) For $f = 0.1$ compute s for $d = 1, 2, 10, 100$. You may use $0.1^{0.5} \approx 0.316$, $0.1^{0.1} \approx 0.794$, $0.1^{0.01} \approx 0.977$. In one or two sentences, what does the $d = 100$ answer mean for the word "neighbourhood" in k-NN?

d	1	2	10	100
s	_____	_____	_____	_____

(c) Lecture 09 showed the fraction of a unit ball's volume within ϵ of its surface is $1 - (1-\epsilon)^d$. For $\epsilon = 0.1$ compute it at $d = 2, 10, 50$ (use $0.9^{10} \approx 0.349$, $0.9^{50} \approx 0.005$). Combine this with the fact that $d_{\max}/d_{\min} \to 1$: why does "the nearest neighbour" stop carrying information as d grows?

Fraction within $\epsilon = 0.1$ of the surface — $d = 2$: _____ $d = 10$: _____ $d = 50$:

(d) A teammate proposes to fix a poorly performing k-NN model on 500 features by "just collecting more data". Using a 10-bins-per-dimension argument (with 10^6 points, how many points per bin at $d = 3$? at $d = 6$? at $d = 12$?), say why that plan fails, and name the two remedies the lecture actually recommends.

Points per bin — $d = 3$: _____ $d = 6$: _____ $d = 12$: _____

Problem 3: Naive Bayes on symptoms, and the SVM margin as geometry

Practices: Bayes-rule classification by counting, Laplace smoothing, and the distance-to-a-hyperplane computation behind support vector machines.

Part I — Naive Bayes. A clinic recorded three symptoms and a lab-confirmed diagnosis for ten patients:

#	Fever	Cough	Headache	Flu
1	Y	Y	Y	Y
2	Y	Y	Y	Y
3	Y	Y	Y	Y
4	Y	Y	N	Y
5	Y	Y	Y	N
6	N	Y	Y	N
7	N	Y	N	N
8	N	N	N	N
9	N	N	N	N
10	N	N	N	N

A new patient presents with **Fever = Y, Cough = N, Headache = Y**.

(a) Estimate the class priors and the six conditional probabilities Naive Bayes needs for this patient, $P(\text{symptom value} \mid \text{Flu})$ and $P(\text{symptom value} \mid \text{no Flu})$, by counting. Compute the Naive Bayes score $P(C)\prod_j P(x_j \mid C)$ for each class and give the prediction. What single number decided it?

class C	$P(C)$	$P(F{=}Y \mid C)$	$P(C{=}N \mid C)$	$P(H{=}Y \mid C)$	score
Flu	_____	_____	_____	_____	_____
no Flu	_____	_____	_____	_____	_____

Prediction: _____

(b) Redo the scores with Laplace smoothing $\alpha = 1$: each conditional becomes $\dfrac{\#\{x_j = v \text{ and class } C\} + \alpha}{\#\{\text{class } C\} + \alpha V_j}$, where $V_j = 2$ is the number of values symptom j can take (leave the priors unsmoothed). Give the smoothed prediction and the smoothed posterior $P(\text{Flu} \mid \mathbf{x})$ = (Flu score) / (sum of both scores). Did smoothing change the prediction, the confidence, or both? Round to 3 decimals.

Smoothed prediction: _____ $P(\text{Flu} \mid \mathbf{x})$: _____

(c) Suppose the clinic added a fourth column, "Chills", that happens to be identical to Fever in every row. Say what Naive Bayes would do to the fever evidence, why that violates the "naive" assumption, and — in one or two sentences — why the classifier often still picks the right class despite such violations.

Part II — SVM margin (self-contained; quizzes will not test SVMs by name, only this geometry). A *linear classifier* in the plane predicts $\text{sign}(\mathbf{w} \cdot \mathbf{x} + b)$, and the distance from a point $\mathbf{x}_0$ to the boundary line $\mathbf{w} \cdot \mathbf{x} + b = 0$ is $|\mathbf{w} \cdot \mathbf{x}_0 + b| \,/\, \|\mathbf{w}\|$. Take $\mathbf{w} = (3, 4)$, $b = -10$, and the six labelled points

point	A	B	C	D	E	F
$\mathbf{x}$	(1, 3)	(2, 4)	(6, 2)	(−1, 2)	(0, 1)	(0, 0)
label y	+1	+1	+1	−1	−1	−1

(d) Compute $\mathbf{w} \cdot \mathbf{x} + b$ and the distance to the boundary for each point. Is every point correctly classified? Which points are closest to the boundary (these are the *support vectors*), and what is the *margin* — the distance from the boundary to the nearest point on each side? A seventh point $(3, 1)$ arrives labelled -1: is it classified correctly, and what is its distance? Finish with one sentence: among all lines that separate the two classes, a support vector machine picks the one with the largest margin — why is that a sensible choice for *new* points? (The full method — how the line is found — is optional module M7.)

point	A	B	C	D	E	F	(3, 1)
$\mathbf{w} \cdot \mathbf{x} + b$	_____	_____	_____	_____	_____	_____	_____
distance	_____	_____	_____	_____	_____	_____	_____

Support vectors: _____ Margin: _____